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Hands-On Modules: Metabarcoding, Metagenomics & Pangenomics

27221 Microbiome Engineering — DTU, MSc Bioengineering

Status: Modules 1 and 2 signed off; notebooks built. Module 3 drafted and proposed, pending time-budget approval from the course coordinator.

Overview

	Module 1 — Metabarcoding	Module 2 — Genome-resolved metagenomics	Module 3 — Pangenomics (<i>proposed</i>)
Duration	2 × 4h sessions	2 × 4h sessions	1 × 4h session
System	Wastewater treatment plant (WWTP) activated sludge	Plant-root synthetic community (SynCom)	Same SynCom isolates as Module 2 (+ optionally its recovered MAGs)
Toolchain	R / DADA2 / phyloseq	anvi'o (+ megahit, fastp for small-scale assembly)	anvi'o (pangenomics workflow)
Environment	JupyterLab + IRkernel, via a shared conda env, accessed through a browser tab inside ThinLinc on the DTU HPC (Gbar/DCC) shared filesystem	Terminal + anvi'o interactive (browser) via ThinLinc on the same shared filesystem	Same as Module 2 — no new environment
Student background assumed	No prior coding proficiency	No prior coding proficiency	Completed Module 2
Deliverable	One figure + short interpretation (per session pair)	One figure + short interpretation (per session pair)	One figure + short interpretation

Student baseline: molecular microbiology + microbial genetics/physiology (per course pre-requisites), **no assumed programming experience**. All notebooks are heavily scaffolded — pre-written code with reasoning-required blanks (>> YOUR DECISION <<), not blank cells.

Environment decision (resolved)

DTU HPC (Gbar/DCC) has no Singularity/Apptainer module and no RStudio Server/Desktop available — confirmed via `module avail` on the actual cluster. RStudio Desktop cannot be installed by students themselves either (no root access, and compute nodes may lack outbound internet). The working setup for **both** R- and anvi'o-based modules is therefore:

- A **single, shared, pre-built conda environment** (not installed per student) staged on the shared filesystem, containing R + DADA2 + phyloseq + vegan + tidyverse + IRkernel + JupyterLab for Module 1, and a separate anvi'o environment (anvi'o's own recommended conda-based install, plus megahit and fastp) for Modules 2 and 3.
- Students activate the relevant environment and, for Module 1, open JupyterLab in a browser tab inside their ThinLinc session; for Modules 2/3, work from a terminal and view anvi'o's interactive interface in a browser tab the same way.
- *(Still open: whether to request Apptainer support from DTU HPC as a longer-term improvement for a nicer RStudio experience in future runs of the course — not blocking for this iteration.)*

Notebook execution model

- **Module 1:** .ipynb notebooks, run live in JupyterLab (R kernel) via ThinLinc — students execute cells directly, not copy-paste. (Originally scoped for RStudio/.Rmd; converted to Jupyter/IRkernel once RStudio was confirmed unavailable on the HPC — see Environment decision above.)
- **Module 2 & 3:** notebooks are structured reference documents (.ipynb with bash cells, plus matching .md) — students copy commands into a terminal session, since anvi'o is CLI-driven. The anvi'o interactive interface is viewed through a browser tab inside the ThinLinc session.

Files produced so far

- **Module 1, Session 1** — M1_S1_reads_to_ASVs.ipynb (+ .Rmd) — live-executable
 - **Module 1, Session 2** — M1_S2_ASVs_to_biology.ipynb (+ .Rmd) — live-executable
 - **Module 2, Session 1** — M2_S1_reads_to_contigsDB.ipynb (+ .md) — copy-paste reference
 - **Module 2, Session 2** — M2_S2_profile_to_MAGs.ipynb (+ .md) — copy-paste reference
 - **Module 3 (proposed)** — M3_S1_pangenomics.ipynb (+ .md) — copy-paste reference
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Module 1 — Metabarcoding: Wastewater Treatment Plant Microbiomes

Dataset

- **Source:** raw 16S rRNA V4 amplicon data from the **Global Water Microbiome Consortium** survey (NCBI SRA, BioProject **PRJNA509305**), the same raw dataset underlying the MiDAS 4 database (Dueholm et al. 2022, *Nat Commun*).
- **Curated subset:** instructor pre-selects ~12-20 samples spanning a few countries and/or process configurations (e.g. conventional activated sludge vs. other configurations), staged on the shared HPC filesystem ahead of class, together with a metadata table (country, plant type/process, and any available performance parameters).
- **Metadata:** sample metadata is in Supplementary Table 1 of Wu et al. 2019 (*Nat Microbiol*), freely downloadable. Real process/environmental variables confirmed available include Continent, Influent BOD, Sludge Age, mean annual air temperature, MLSS, DO, F/M ratio, and nutrient removal efficiencies — good candidate “engineering variables”

for Session 2. (You're finalizing the exact sample subset and column names directly from the file.)

- **Reference taxonomy: MiDAS 4** (attribution: Aalborg University / Center for Microbial Communities), used at the DADA2 taxonomy-assignment step — gives genus/species-level resolution that generic universal databases don't achieve well for WWTP bacteria.
- **Why this dataset:** real, globally-sourced, well-documented, metadata-rich, and directly on-topic for a biotechnology/engineering audience — students can ask “does the community track with how the plant is run?” with real data.

Learning objectives targeted

- Describe structure/function/dynamics of an engineered-environment microbiome.
- Describe and evaluate sequencing-based microbiome characterization methods.
- Analyze/interpret microbiome data using computational and statistical approaches.

Session 1 (4h) — Raw reads → ASVs

Time	Segment
0:00–1:00	Theory: amplicon sequencing logic, primer choice/regions, sequencing error models, why ASVs (DADA2) supersede OTU clustering.
1:00–1:15	Break
1:15–3:30	Hands-on (JupyterLab, live execution): import demultiplexed FASTQs → quality profiling (plotQualityProfile) → filter/trim (guided reasoning on truncation length from quality plots) → error learning (learnErrors) → denoise (dada) → merge pairs → build sequence table → chimera removal → final ASV table. Every cell pre-written; blanks require a parameter choice justified from the preceding plot/output, not free coding.
3:30–4:00	Wrap-up: what does a row/column of an ASV table represent; sanity checks (read-tracking table through the pipeline).

Session 2 (4h) — ASVs → biology

Time	Segment
0:00–0:45	Theory: taxonomy assignment logic, alpha vs. beta diversity, ordination, what a “differential abundance” claim actually means (and its pitfalls).
0:45–1:00	Break

Time	Segment
1:00–3:00	Hands-on: assign taxonomy against MiDAS 4 → build phyloseq object with sample metadata → alpha diversity comparison across an engineering variable (e.g. plant/process type, sludge age, F/M ratio) → ordination (e.g. NMDS/PCoA on Bray-Curtis) → one simple differential abundance comparison.
3:00–3:45	Students produce their deliverable figure + draft interpretation.
3:45–4:00	Discussion / share-out of a few figures.

Deliverable

One figure (diversity plot, ordination, or differential abundance plot) + a short paragraph: what shifts with the chosen engineering variable, and what that would mean for someone operating the plant.

Module 2 — Genome-Resolved Metagenomics: Plant-Root Synthetic Community

Dataset

- **Primary:** Selten et al. — shotgun metagenomics of a complex, defined plant-root SynCom (NCBI SRA BioProject **PRJNA1131994**), with known constituent isolate genomes serving as **ground truth** for bin recovery/validation — pedagogically valuable since students can check “did we get back what was put in?”
- **Bonus/bridge exercise:** the same samples were also sequenced by 16S amplicon (BioProject **PRJNA1191388**) — optional short comparison at the end of Session 2, contrasting what Module 1’s amplicon approach would show vs. genome-resolved metagenomics on the *same* system.
- **Framing story:** open Session 1’s theory block with Carrión et al. 2019 (*Science*) — the 2-strain sugarbeet SynCom where metagenomics pinned a biocontrol mechanism to specific gene clusters — as a compact, well-known illustration of *why* genome-resolved metagenomics matters for engineered communities, before moving to the more complex dataset used for hands-on work.

Learning objectives targeted

- Describe/evaluate microbiome characterization methods (sequencing-based, model systems).
- Explain synthetic biology / systems biology advances relevant to microbiome engineering.
- Design strategies to manipulate microbiomes (ties to “designing microbial communities”).
- Analyze/interpret microbiome data computationally.

Session 1 (4h) — Small-scale, live: reads → contigs database

Time	Segment
0:00–1:00	Theory: shotgun vs. amplicon, assembly basics, what a contigs database is, the Carrión et al. framing story, why this matters for engineered/designed communities.
1:00–1:15	Break
1:15–3:30	Hands-on (terminal via ThinLinc, on a small pre-sampled read set — small enough to finish live): QC (fastp) → quick assembly (megahit) → anvi-gen-contigs-database → anvi-run-hmms → gene calling / basic functional annotation.
3:30–4:00	Wrap-up: first look at a contigs database and the anvi-interactive interface (opened in browser via ThinLinc).

Session 2 (4h) — Precomputed profile → genome-resolved exploration

Time	Segment
0:00–0:45	Theory: read recruitment/mapping concept, binning logic, MAG quality metrics (completion/redundancy).
0:45–1:00	Break
1:00–3:00	Hands-on: explore a pre-built, full-scale merged profile (too large to generate live) in anvi-interactive → binning → pull out a bin of interest → compare against known isolate genomes (ground truth, via ANI) → basic functional/taxonomic characterization of the recovered bin.
3:00–3:30	<i>(Optional bridge):</i> brief comparison against the matched 16S amplicon data (PRJNA1191388) for the same samples — what would Module 1’s approach have told us, vs. what genome-resolved metagenomics reveals?
3:30–4:00	Students produce their deliverable figure + draft interpretation; share-out.

Deliverable

One figure (e.g. annotated bin overview, or an anvi-interactive/binning screenshot) + a short paragraph: which community members were recovered, one functional or taxonomic claim about them, and what engineering decision it could inform (e.g. relevant to the SynCom’s design goal).

Module 3 — Pangenomics (*proposed*)

Status: proposed, pending time-budget approval from the course coordinator. Designed as a direct continuation of Module 2 — same dataset, same tool, no new environment to build.

Dataset

Same SynCom isolate genomes already staged for Module 2's ground-truth comparison step, reused directly as the pangenome input — no new data-staging effort. Optionally, the MAGs students recovered in Module 2 can be added as “internal genomes” alongside the isolates, to ask whether a recovered MAG sits where its matched isolate does in gene-content space.

Learning objectives targeted

- Design strategies to manipulate/design microbial communities (redundancy vs. complementarity is a direct design question for a SynCom).
- Explain synthetic biology / systems biology advances relevant to microbiome engineering.
- Analyze/interpret microbiome data computationally.

Session (4h) — Core, accessory, and the redundancy question

Time	Segment
0:00-0:45	Theory: what a pangenome is (core vs. accessory vs. singleton gene clusters), how <i>anvi'o</i> builds one (protein clustering via MCL on translated ORFs), and why this matters for a <i>designed</i> community — functional redundancy vs. complementarity across strains is a direct SynCom design question (“do we need all of these members?”).
0:45-1:00	Break
1:00-3:00	Hands-on: build a genomes storage from the isolate genomes → run the pangenome analysis → explore interactively (<i>anvi-display-pan</i>) → extract core/accessory/singleton counts per genome → functional enrichment of the accessory genome (reusing the COG annotation from Module 2).
3:00-3:45	Students produce their deliverable figure + draft interpretation.
3:45-4:00	Discussion / share-out.

Deliverable

One figure (e.g. an annotated *anvi-display-pan* screenshot, or a bar plot of core/accessory/singleton gene cluster counts per isolate) + a short paragraph: which isolate(s) look most functionally redundant with others in the SynCom, which look most distinct, and what that would mean for a decision to simplify the community (drop a strain) or not.

Logistics checklist (pre-class, instructor-side)

- ☐ Confirm ThinLinc access + shared filesystem paths for all enrolled students.
- ☐ Build and stage a **shared, read-only conda environment** for Module 1 (R + DADA2 + phyloseq + vegan + tidyverse + IRkernel + JupyterLab) — all students activate the same env rather than building their own.
- ☐ Build and stage a **shared anvi'o conda environment** for Modules 2/3 (anvi'o's own recommended install, plus fastp and megahit).
- ☐ Module 1: finalize the ~12-20 sample WWTP subset from PRJNA509305 + the confirmed metadata columns from Supplementary Table 1; stage MiDAS 4 reference taxonomy files.
- ☐ Module 2: select and stage a small subsampled read set from PRJNA1131994 for live Session 1 use; pre-build the full merged profile database for Session 2 (too heavy to build live); stage known isolate reference genomes and the `external-genomes.txt` manifest for ground-truth comparison; stage matched amplicon data (PRJNA1191388) if using the bonus bridge exercise.
- ☐ Module 3 (*if approved*): no new staging needed beyond what Module 2 already requires — reuses the same isolate genomes and `external-genomes.txt`.
- ☐ Confirm anvi'o's local web server for `anvi-interactive/anvi-display-pan` and JupyterLab are both reachable from within a ThinLinc browser session (port/proxy considerations on the HPC).
- ☐ Sanity-check exact anvi'o CLI flags used in the Module 2/3 notebooks against whatever anvi'o version is actually installed (command syntax has changed across versions).
- ☐ Decide on final deliverable submission mechanism (shared filesystem folder, LMS upload, etc.).

Next steps

1. Course coordinator decision on Module 3's time allocation.
2. You finalize and stage real data for Module 1 (WWTP subset + metadata) and Modules 2/3 (SynCom subsamples, precomputed profile, isolate genomes) on the shared filesystem.
3. Build the two shared conda environments (R/Jupyter for Module 1; anvi'o for Modules 2/3) — happy to draft the `environment.yml/install` steps for either whenever useful.
4. Once real paths exist, update the placeholder config values in each notebook (`raw_reads_dir`, `CONTIGS_DB`, `PROFILE_DB`, etc.) to match.
5. Optional, longer-term: ask DTU HPC whether Apptainer/Singularity support could be added, to enable a smoother RStudio Server setup in future runs.